

Two Ways a Design Fails: When Does a Synthetic LLM Panel See the Danger?

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Abstract

Deploying an AI-advice interface can fail in (at least) two ways. It can make human reliance *heterogeneous and user-sensitive*—safe on average, dangerous for particular people (an *over-dispersion* failure). Or it can make users *over-rely on a confidently wrong AI* under a coercive framing (a *dark-pattern* failure). Catching either normally requires an expensive human study. A tempting shortcut is to screen an interface *before* that study with a calibrated multi-agent LLM panel that plays a synthetic cohort of users. We build such a panel—two-axis (over-dispersion; wrong-advice adoption), conflict-conditioned, preregistered—and ask a question prior to “does it predict humans?”: *are its danger signals even stable across the LLMs that power it?* They are not. The signals are strongly **model-dependent**. Using a matched contrast that holds the wrong advice constant and varies only the framing, we find a real *bundled coercive-framing effect* (item-clustered odds ratio 1.90 on one dataset, 1.17 on the other; a non-coercive placebo stays at or below the neutral baseline)—but it is model-dependent: a frontier model (gpt-5.5) shows the smallest framing effect and the lowest absolute wrong-advice adoption, while smaller and mid-tier models adopt more, idiosyncratically. **The choice of panel model is therefore a first-order, under-appreciated methodological decision: a single-frontier-model panel can suppress a warning that other panel models raise.** A secondary, backend-stable manipulation check shows the panel faithfully propagates a prompted trust disposition: an explicitly trusting-novice persona is the highest-adopting persona in all 12 model×domain cells (item-clustered odds ratio ≈15). We contribute (i) the model-dependence finding—including the model-dependent coercive-framing effect—as direct evidence for the silicon-sampling reliability debate, and (ii) the two-axis, over-dispersion-as-safety protocol on a preregistered, audit-gated pipeline. We do *not* claim to predict individual humans or a validated deployment screen, and we are explicit that the persona result is largely a property of the prompt; we characterize *when a synthetic panel does and does not see the danger*, and design the human study that would ground it.

CCS Concepts

• Human-centered computing → HCI design and evaluation methods; • Computing methodologies → Artificial intelligence.

Keywords

human-AI decision making, over-reliance, dark patterns, LLM simulation, synthetic users, pre-deployment evaluation, silicon sampling

1 Introduction

Consider a product team about to ship a review-triage tool. The interface shows an AI’s verdict and a confidence score and—to feel trustworthy—a highlighted “explanation.” Averaged over a pilot, the numbers look fine: accuracy is up, users seem to like it. What the

average hides is *who* the interface endangers and *when*. A cautious domain expert barely moves; a *trusting novice* takes the AI’s word almost every time—and if a coercive variant presents a confidently wrong verdict, that same novice follows it off a cliff while the aggregate barely twitches. The design has not failed on average. It has failed for a person, in a way the mean cannot see.

This is the gap between *mean accuracy* and *safety*. Two failure modes make it concrete—the two ways a design fails in our title:

- **Axis 1—over-dispersion.** Reliance becomes *heterogeneous across users* in a way the interface induces. The team ships something safe for the median user and unsafe for the tails. Variance, usually a nuisance to be averaged away [3, 7], is here the *safety quantity of interest*.
- **Axis 2—wrong-advice adoption under coercion.** Under a coercive framing, users *systematically adopt a wrong AI*. This is dangerous even at zero dispersion: everyone can fail together [4, 10, 16].

The standard way to detect either failure is a human-subjects study—powered, IRB-approved, and slow. Teams iterate on interfaces far faster than they can run such studies. This paper asks whether a *cheap pre-deployment screen* can act like a smoke alarm: high-recall, run in minutes, flagging interfaces that *warrant* a human study and clearing those that plainly do not. Not a replacement for the human study—a *triage* in front of it.

The screen we investigate is a **calibrated synthetic panel**: persona-conditioned LLM agents that role-play a cohort of users on the interface, from which we read the two axes directly. Synthetic users are having a moment in HCI [1, 2, 12], and so is the backlash: a growing literature warns that LLM-simulated people are *unreliable proxies*—they collapse human diversity, echo the base model’s priors, and mis-estimate distributions [5, 6, 8, 14]. We take that critique seriously enough to ask a question that comes *before* “does the panel predict humans?”: **is the panel’s danger signal even stable across the LLMs you might build it from?** If not, then any claim that “the panel sees the risk” is really a claim about a particular model—and swapping the model can erase the warning.

Our central finding is that the danger signal is **strongly model-dependent**, and not in a tidy way (Figure 1). Under a coercive, high-authority framing that presents a wrong AI verdict, the *frontier* model we test (gpt-5.5) shows the smallest framing effect and the lowest absolute wrong-advice adoption, while smaller and mid-tier models adopt more, but **idiosyncratically**: the smallest model is at chance, one mid-tier model (gpt-4.1) is the only one with above-chance adoption, and two independent-vendor frontier models are at chance. Using a matched contrast that holds the wrong advice constant and varies only the framing (Section 5.3), the coercive framing raises wrong-advice adoption significantly on one dataset (item-clustered odds ratio 1.90) and modestly on the other (1.17), with a non-coercive placebo at or below the neutral baseline—but

again the frontier model barely moves. We interpret the association with capability cautiously—alignment, reasoning behavior, or training could drive it as much as raw capability—but the practical upshot is robust: **the model you choose to power a synthetic panel is a first-order methodological decision.** A team that screens with a single frontier model may see no warning because *that model* does not follow a coercively-framed wrong AI that its less-capable human users might—a risk-masking failure mode we demonstrate *within the panel* and hypothesize (but do not show) for humans.

A secondary result is a backend-stable manipulation check (Figure 3). Even where the model mean shifts, every model ranks the *same* persona highest: an **explicitly trusting-novice** persona is the highest-adopting persona in **12 of 12** model $\times$ domain cells (item-clustered odds ratio ≈ 15). We read this honestly: the persona is *prompt-defined* to be trusting, so this is substantially a demonstration that the panel faithfully and stably propagates a prompted disposition into behavior across backends—a useful reliability property, but *not* evidence that the panel discovers a latent vulnerable human population. Whether the ordering survives removing the explicit “trusting” and “novice” cues, and whether it maps to real people, are questions for descriptor ablation and the human study.

We make two contributions, and we are explicit about their limits:

- (1) **Model-dependence of synthetic-panel screening signals**—direct evidence for the silicon-sampling reliability debate: the *same* preregistered protocol yields strong wrong-advice adoption and a significant coercive-framing effect on smaller/mid-tier models, both attenuated by gpt-5.5, replicated across two review datasets. Panel-model choice is a first-order methodological decision that can, within the panel, suppress a warning other models raise.
- (2) **The two-axis protocol**—over-dispersion-as-safety plus wrong-advice adoption, conflict-conditioned and preregistered—proposed as a reusable *testbed* for reasoning about interface-level reliance safety (not a validated screen), built on a preregistered, audit-gated pipeline whose process integrity (Section 4) is part of the method.

The persona result above is reported as a supporting manipulation check, not a standalone contribution.

We do *not* claim a validated, model-agnostic detector, and we do *not* claim to predict individual humans. Our panel $\leftrightarrow$ human correspondence is null in every configuration we can currently test (all underpowered). We frame this as an open validation gap and describe the human study (Section 8) designed to close it.

2 Related Work

Simulated users, and the reliability critique. LLMs as stand-ins for human participants have been proposed for surveys, user studies, and formative evaluation [1, 2, 12]. A parallel literature documents their failure as proxies: they compress opinion diversity and over-represent majority views [14], mis-calibrate distributions [5], and caricature rather than simulate subpopulations [6, 8]. We sit inside this debate but change the target: we do not ask the panel to *predict* a human distribution, we ask it to *triage* an interface, and we isolate

a specific, measurable pathology—*model-dependence of the triage signal*—that the prediction framing obscures.

Reliance and over-reliance on AI advice. A large body of work studies when people appropriately rely on AI given confidence and explanations [3, 4, 9, 15, 17], often finding explanations *increase* over-reliance rather than calibrate it [3, 16]. This work treats between-user variance as noise around an average. We invert that: reliance *heterogeneity* is our axis-1 safety DV, and coercive wrong-AI adoption is axis-2.

Dark patterns and coercive interfaces. Deceptive and coercive designs are well catalogued [10, 11]; our axis-2 gives a computational signal for one coercive mode (a confidently wrong AI) and, via personas, a read on *who* it captures.

Offline evaluation of human-AI teams. Closest is offline estimation of team performance for decision support [13]; that work studies bias and complementarity, not interface-level safety triage. The gap we fill is a *pre-study screen for interface reliance safety*, and—critically—a characterization of when such a screen’s signal is trustworthy.

3 The Two-Axis Triage: Definitions

We evaluate an interface i on two axes, both measured on the panel’s decisions.

Axis 1—between-user reliance over-dispersion. For each item, personas produce reliance decisions; we estimate the *excess variance* beyond the binomial $p(1-p)$ a single homogeneous user would produce (a beta-binomial over-dispersion estimator, parameter ρ). We compute it *conflict-conditioned*: only on trials where the persona’s independent System-1 judgment disagrees with the shown AI, isolating genuine reliance from trivial agreement. As a stable, bounded summary that avoids the beta-binomial estimator’s boundary behaviour at extreme rates (Section 4), we report *mean panel disagreement*—the mean across items of the per-item variance of the six personas’ reliance decisions—which tracks ρ monotonically and is what we plot; we report ρ alongside it. High axis-1 means the interface makes reliance user-sensitive.

Axis 2—wrong-advice adoption under coercion. Under a coercive “dark” framing, the interface shows a *guaranteed-wrong* AI verdict (the negation of ground truth) at high confidence with authority and accountability cues. Axis-2 is the rate at which personas adopt it. To separate the *framing* from ordinary wrong-advice following, we contrast it against neutral and placebo framings on matched items (Section 5.3). It is dangerous even at zero dispersion—a uniform failure.

The triage rule (design, not yet validated). We learn thresholds $\tau_{\text{disp}}, \eta_{\text{level}}$ once in an atomic UI feature space; either axis over threshold $\Rightarrow$ *recommend a human study*; both low $\Rightarrow$ *release candidate*; out-of-distribution or high-uncertainty $\Rightarrow$ *abstain*. The rule is deliberately high-recall (a missed dangerous interface is worse than a false alarm). We report it as a *design*; its validation depends on the human study (Section 8), and thresholds remain unfrozen pending that study.

4 Method: The Calibrated Synthetic Panel

Panel = personas × models. Each agent is a persona (a vector of domain skill, AI literacy, risk sensitivity, caution) crossed with a base LLM. Personas span novice/expert × trusting/skeptical plus two moderates (six in total).

Dual-system decisions. Each agent first produces a **System-1** judgment *without* AI (a frozen, no-AI anchor), then a **System-2** decision *with* the AI advice and framing. Reliance is defined against the System-1 anchor—what makes conflict-conditioning possible and prevents the “everyone-agrees-so-everyone-complies” collapse of a naive panel.

Textual prompt encoding of the interface. Agents receive a *textual serialization* of each interface condition—not screenshots and not a vision model. The AI prediction, its confidence, and (for the explanation conditions) the highlighted tokens are rendered as text in the prompt, following the Bansal et al. [3] condition definitions (confidence-only; single/double highlighted-token explanations; adaptive and expert-adaptive). The dark condition negates ground truth and adds high confidence plus authority/accountability language; a placebo condition presents the real AI non-coercively. This means we study *prompt-level* interface framing; pixel-level layout, visual salience, and typography are not modelled, and a multimodal rendering is future work. The exact per-model prompt templates are in the supplement.

Stimuli and datasets. Items are drawn from the Bansal sentiment benchmark, held out from any threshold fitting, and selected to concentrate on loci of human reliance heterogeneity (disclosed; human outcomes are used only for *item selection*, never leaked into agent prompts—item selection ≠ label leakage). We run two *review-sentiment datasets*: **beer reviews** and **Amazon book reviews (amzbook)**. Both are English binary sentiment; we therefore treat this as replication across two datasets within one task family, not broad cross-domain generalization (Section 7).

Models. A within-provider capability ladder (gpt-4o-mini → gpt-4.1 → gpt-4o → gpt-5.5, identical config) plus independent-vendor frontier models (claude-sonnet-4.5, gemini-2.5-pro): six models total.

Rigor spine and process integrity. The analysis was preregistered and frozen with UTC timestamps before results; correspondence analyses were coded blind; every merge was gated on an independent audit *and* a from-raw numeric re-derivation. This spine caught three silent-corruption bugs that would each have produced a plausible-but-false result: an axis-2 *sign inversion* (scoring adoption against ground_truth instead of the displayed $1 - \text{ground_truth}$), a beta-binomial *boundary artifact*, and a decision parser that silently *fabricated* a (decision=0, conf=0.5) sentinel on parse failure. We report these openly: for a paper whose headline includes a negative and a nuanced positive, a pipeline that catches its own errors is what makes the numbers trustworthy.

5 Experiments and Results

We report per-model (no pooling across incomparable models), with effect sizes, confidence intervals, and Benjamini–Hochberg (BH)

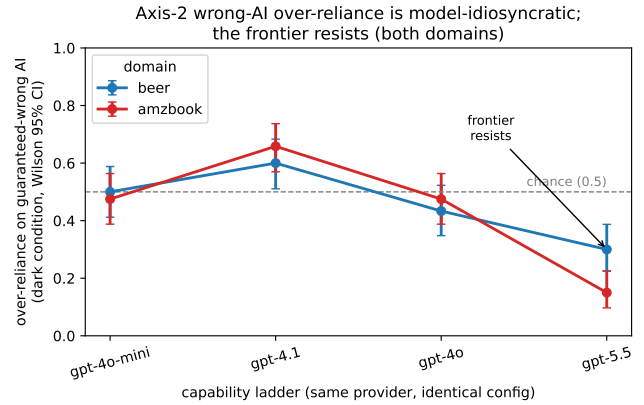


Figure 1: Axis-2 wrong-advice adoption on the guaranteed-wrong interface across the same-provider capability ladder, both datasets (Wilson 95% CIs; dashed line = 0.5). Adoption is non-monotonic and idiosyncratic; the frontier model (gpt-5.5) has the lowest adoption on both datasets. A non-arbitrary baseline for “resists” is given by the matched framing contrast (Section 5.3).

correction across the per-model test family. There are $n = 120$ dark-condition trials per model×domain cell (6 personas × 20 items).

5.1 Real human over-dispersion exists (anchor)

On the Bansal human data, conflict-conditioned reliance over-dispersion is real and excludes zero ($\rho \approx 0.067$); no-AI reliance behaves as a stable trait (≈ 0.74) while AI-assisted reliance is a user×task interaction (0.32–0.41). This anchors axis-1 as a measurable human quantity—and, honestly, does not generalize to a second human dataset (Lu & Yin), which we report as a boundary.

5.2 The danger signal is model-dependent

On the guaranteed-wrong interface, axis-2 wrong-advice adoption is *not* a clean function of capability (Figure 1). Across the ladder (beer / amzbook): gpt-4o-mini 0.50 / 0.48, gpt-4.1 0.60 / **0.66**, gpt-4o 0.43 / 0.48, gpt-5.5 **0.30** / **0.15**. The adoption peak at gpt-4.1 is real (between-model Fisher vs. gpt-4o $p = 0.014/0.006$) but **model-idiosyncratic**: the smallest ladder model is at chance, and after BH correction gpt-4.1’s elevated adoption survives on amzbook ($p = 0.003$) but is only nominal on beer (uncorrected $p = 0.035$; BH $p = 0.11$). Independent-vendor frontier models are at chance (claude 0.49, gemini 0.53 on beer; 0/3 vendors above chance). Rather than interpret adoption against a 0.5 “chance” baseline—which is not a natural no-effect point, since personas are not random guessers—we ground the effect in a matched framing contrast below.

Axis-1 tells a parallel story (Figure 2). Mean panel disagreement across the ladder is 0.11 / 0.16 / 0.10 / **0.012** (beer) and 0.12 / 0.19 / 0.11 / **0.015** (amzbook): gpt-5.5 collapses to near-homogeneity on both datasets, and the beta-binomial ρ (which mean disagreement tracks) collapses in step (e.g. ρ from 0.32–0.58 down to ≈ 0 at the frontier). Axis-1 is computed over the full 5-condition design (5 conditions × 6 personas × 20 items = 600 trials/model), distinct from

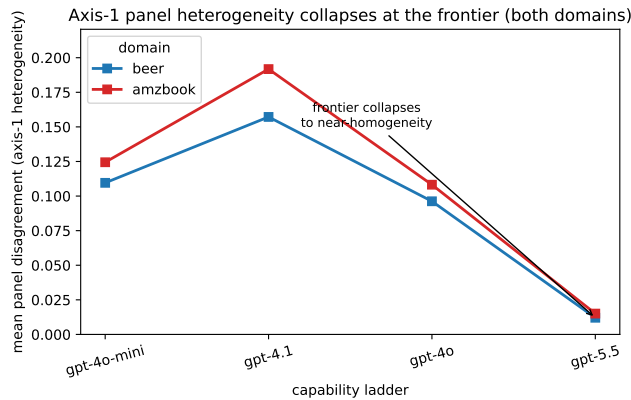


Figure 2: Axis-1 mean panel disagreement across the ladder, both datasets. Heterogeneity collapses at the frontier (gpt-5.5) on both.

the 120 dark-only axis-2 trials; the collapse is not a no-conflict artifact, since genuine conflict trials remain present at the frontier (e.g. 215/600 on amzbook). Because conflict-conditioning yields slightly different effective subsets per model, we verified the collapse also holds on the common conflict subset shared across models.

Reading. gpt-5.5’s cross-domain low adoption and axis-1 collapse are the most consistent patterns; adoption magnitude is otherwise model-specific. A synthetic panel’s aggregate reading is therefore a property of the panel’s model as much as of the interface: swapping in the frontier model can turn a warning into no warning *within the panel*.

5.3 Is it coercion, or just wrong-advice following? A matched framing contrast

Adoption of a guaranteed-wrong AI could reflect ordinary wrong-advice following rather than the *coercive framing* (high confidence, expert-authority and accountability cues). To separate them we use a within-item contrast: on the subset of items where the AI advice is *naturally wrong* under the neutral condition (8 items/cell), the same wrong label is shown under neutral, non-coercive placebo, and coercive-dark framings—so any difference in adoption is attributable to the framing, not to the advice being wrong. We fit a logistic model $\text{adopt} \sim \text{dark} + \text{placebo} + \text{model}$, clustered by item.

The coercive framing significantly increases wrong-advice adoption on beer (odds ratio **1.90**, 95% CI 1.42–2.53, $p = 1.3 \times 10^{-5}$) and modestly on amzbook (OR **1.17**, $p = 0.01$); the non-coercive placebo stays at or below the neutral baseline (beer OR 0.94, n.s.; amzbook OR 0.88, $p = 0.005$, a small *protective* effect), confirming the effect is the coercive framing and not mere AI-presence. The effect is clearest on the conflict-conditioned switch rate—the fraction of trials where a persona whose System-1 judgment was *correct* switches to the wrong advice: coercive framing roughly doubles-to-triples it for non-frontier models (e.g. beer gpt-4.1 $0.19 \rightarrow 0.51$; gpt-4o $0.12 \rightarrow 0.29$), while the frontier gpt-5.5 barely moves ($0.04 \rightarrow 0.10$). This licenses two honest statements: axis-2 reflects a real *bundled*

coercive-framing effect (we cannot separate confidence from authority from accountability, which co-vary in the dark condition), and that effect is model-dependent—smallest at the frontier. It also replaces the arbitrary 0.5 baseline: “gpt-5.5 resists” means it shows the smallest matched framing effect, not merely adoption below one-half.

5.4 One signal is robust: which persona the panel ranks highest

5.5 A backend-stable manipulation check: the panel propagates a prompted trust disposition

The aggregate adoption *rate* is model-idiosyncratic (range 0.30 across beer models, 0.51 across amzbook models). The *ordering* of which persona adopts most is stable (Figure 3). An **explicitly trusting-novice** persona is the highest-adopting persona in **all 12** model×domain cells. Because the 20 items are reused across personas and the 12 cells share prompts, items, and providers, we do not treat trials as independent: clustering by item (the main pseudo-replication unit), a logistic model gives an odds ratio of **14.9** (95% CI 10.9–20.2) for that persona relative to the others, and an item-level bootstrap of the mean per-cell adoption gap gives +0.53 (95% CI 0.48–0.57). The effect does not collapse under item clustering.

We interpret this carefully, and *demote it from a headline finding to a manipulation check*. The persona is *prompt-defined* to be trusting and novice, so “it adopts the AI most” is substantially a consequence of the instruction, not a discovery. What the result does show is a genuine reliability property: the panel *faithfully and stably* translates a prompted disposition into behaviour, and the resulting ordering is consistent across six backends and two datasets (independent-vendor per-persona profiles correlate at mean pairwise Spearman 0.87 on beer; Figure 4). It is *not* evidence that the panel discovers a latent vulnerable human population; a chance baseline of 1/6 per persona would be inappropriate here, since the personas are not exchangeable (this one was constructed to be the most trusting). Whether the ordering survives removing the explicit “trusting”/“novice” descriptors (a descriptor ablation), and whether it corresponds to real users, are questions we leave to future work and the human study.

5.6 Panel↔human correspondence is unvalidated (honest null)

Across every configuration we can test, the panel’s per-condition axis-1 signal does *not* significantly track the human anchor’s (Spearman on $n = 5$ conditions; null under BH for all six models). On amzbook the correspondence is *degenerate*—the human anchor’s per-condition over-dispersion is near-constant, so the test is uninformative rather than a genuine null. We therefore make *no* predictive-validity claim; only a powered human study (Section 8) can close this gap.

6 Discussion

Panel-model choice has screening consequences. The field’s instinct is that a *better* base model makes a *better* simulator. For

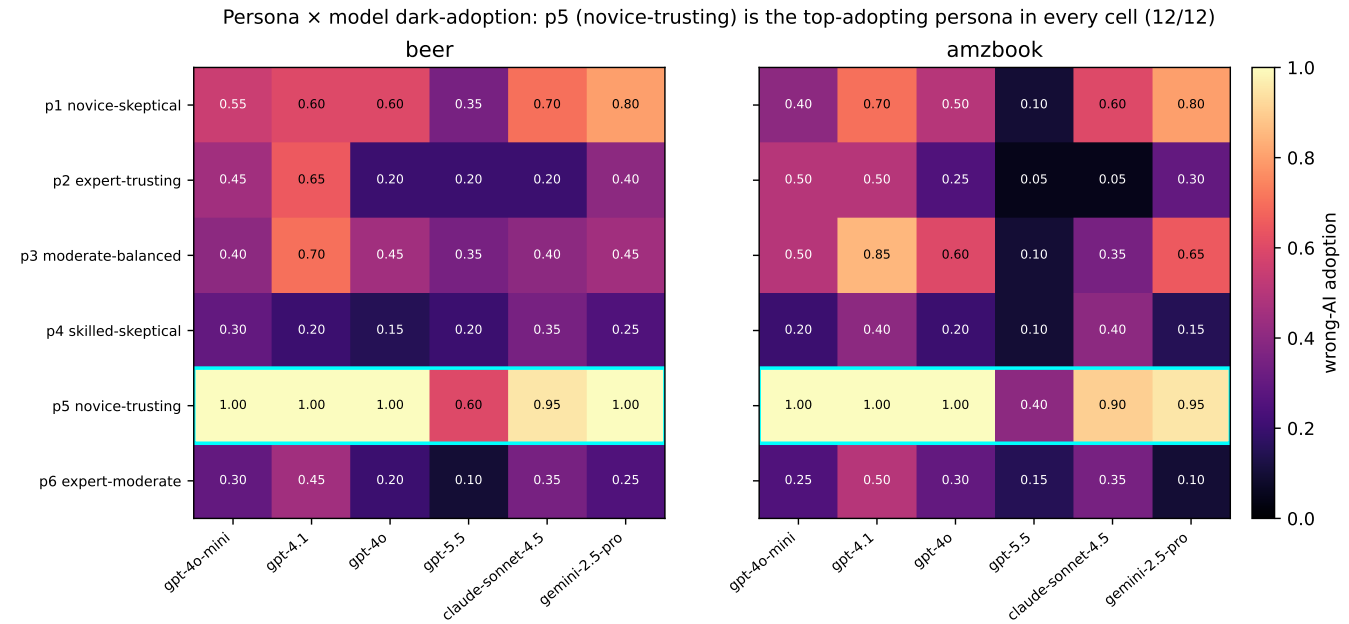


Figure 3: Persona × model wrong-advice adoption on the dark condition, both domains. The trusting-novice persona (p5, highlighted) is the single highest-adopting persona in all 12 model×domain cells.

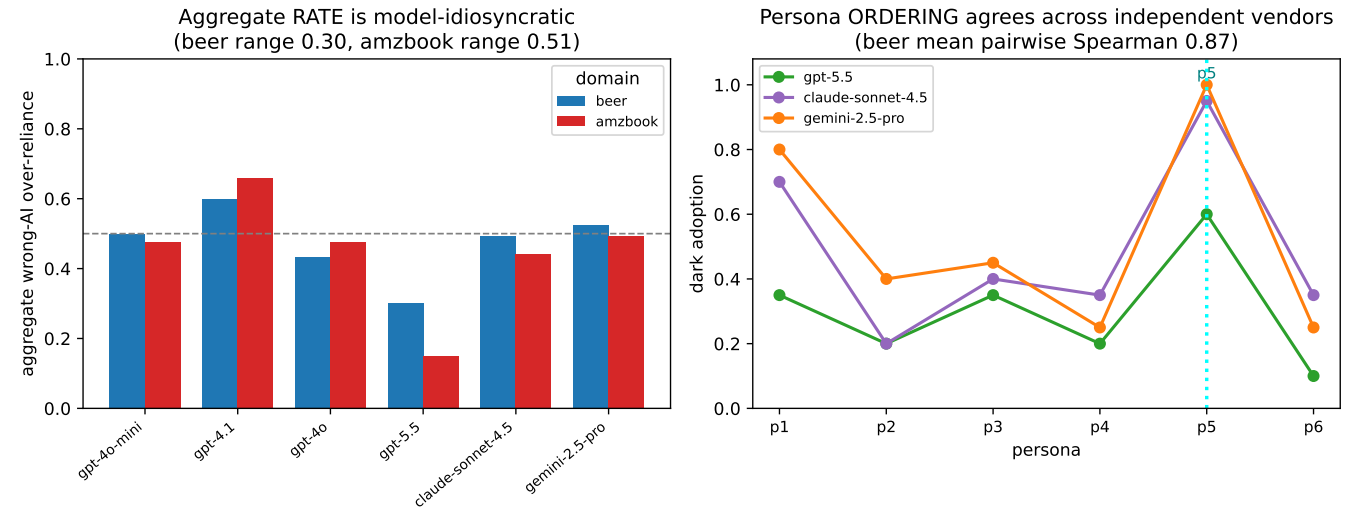


Figure 4: Left: aggregate wrong-advice adoption is model-idiosyncratic (large spread across models). Right: the per-persona adoption profile is consistent across independent vendors (gpt-5.5, claude, gemini), all peaking at the prompted trusting-novice persona (mean pairwise Spearman 0.87 on beer). This is a reliability property of persona-prompt propagation, not population discovery.

danger-screening, the opposite can hold: the frontier model we test follows a coercively-framed wrong AI least, so a single-frontier-model panel can under-report a risk *within the panel*. If—a hypothesis for the human study, not a demonstrated fact—the frontier model is less human-like than a team’s actual users, that under-report would carry to deployment. Either way, “which LLM should

power my synthetic study?” becomes a first-order methodological choice, arguing for **deliberately including lower-tier models** in a screening panel.

Read persona output as a prompt property, cautiously. The panel stably ranks a prompted trusting-novice profile highest, but because

that profile is defined to be trusting, this is a reliability property of the simulation, not a discovery of who is at risk. At most it can *generate hypotheses* (“this interface may especially affect users who resemble a trusting novice”) for a human study to test—not an absolute or validated risk gauge.

A concrete stance in the silicon-sampling debate. Rather than a blanket “LLMs (don’t) simulate people,” we offer a measured, replicated claim: *aggregate* wrong-advice and coercive-framing signals are model-dependent and do not transfer across models, while a prompted *relative* persona ordering is propagated stably. That distinction is actionable for anyone building simulated-user tools.

7 Limitations and Threats to Validity

- **Synthetic personas are not people**—the central threat. The model-dependence result makes it concrete: “the panel” is not one thing, so any human-facing claim must name the model. Only the human study (Section 8) grounds *which* model, if any, tracks people.
- **The screen is not model-agnostic.** Our evidence shows the aggregate signal does not transfer across capability tiers; we do not claim a deployable, model-agnostic detector.
- **Correspondence is unvalidated and underpowered** ($n = 5$ conditions; amzbook degenerate). No predictive-validity claim is made.
- **Two review datasets, both binary sentiment.** Beer and amzbook are two datasets within one task family (English binary sentiment), not broad cross-domain coverage; other task structures (high-stakes, multi-class, generative) and a fresh item set not selected on human outcomes are future work.
- **Coercion is a bundled manipulation.** The dark condition varies confidence, expert authority, and accountability cues together, so the framing effect (Section 5.3) is a *bundled* coercive effect; we do not factorially separate its components.
- **Single generation per prompt.** Each agent decision is one draw at the persona’s temperature; we do not estimate generation stochasticity or prompt-paraphrase sensitivity, and inference is limited to these fixed personas, models, and datasets rather than sampled populations.
- **Item selection uses human reliance variance** (disclosed). Holding items out from threshold fitting does not remove selection on the human criterion; a fresh-item human check remains, so we treat this as unresolved pending the human study.
- **Prompt-level, not pixel-level.** Agents see a textual encoding of the interface, so we study prompt framing rather than visual interface effects; a multimodal rendering is future work.

8 Planned Human Validation

This study is designed but not yet run or preregistered. The one thing synthetic compute cannot buy is grounding against real people. We have designed a stripped, axis-2-only human study (within-subjects, $N \approx 40$, Prolific) that would test the paper’s two strongest,

cross-domain-robust positives against humans: (H1) a guaranteed-wrong AI interface elicits more wrong-advice adoption than a faithful/placebo baseline; (H2) a *trusting-novice human index* (high self-reported AI-deference $\times$ low domain skill) predicts the highest wrong-AI adoption—the human test of Section 5.2; and (H3) *which* ladder model’s panel best tracks the human pattern, with the sharp prediction that a mid-tier model tracks humans better than the resistant frontier—i.e. the most-capable panel model may be the least human-faithful screener. All outcomes, including nulls, would be publishable bounds on synthetic-panel validity.

9 Conclusion

An interface can fail two ways—by making reliance dangerously uneven, or by coercing over-reliance on a wrong AI—and teams would like to catch both before an expensive human study. We built a calibrated synthetic panel to do that triage and found that its danger signal is **model-dependent**: a matched framing contrast shows a real but model-dependent coercive-framing effect, smallest at the frontier model we test, so the choice of panel model is itself a methodological decision with screening consequences. Beneath that instability the panel at least propagates a prompted disposition stably—it ranks an explicitly trusting-novice persona highest across every model and dataset—but we read that as a reliability property of the simulation, not a discovery of who is at risk. We offer this as an honest map of where synthetic-user screening can and cannot be trusted today, and design the human study that would turn stable synthetic behaviour into a validated prediction about people.

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